Traffic and Crowd Monitoring - Backend Component View

Web Server [Container]
Nginx / reverse proxy

HTTP: /api/ requests

Application API Container - one FastAPI / Uvicorn process

API Router Layer
[Components]
FastAPI routers
Uploads / results / history / assets

Analyse image

Submit / read progress

Query results / assets

Image Analysis
[Component]
Python / OpenCV
Validate and process JPEG/PNG

Video Analysis [Component]
Python / ThreadPoolExecutor
Worker and frame orchestration

Detect image

Persist results

Detect sampled frames

Persist jobs / results

Shared Detection
[Components]
OpenCV / YOLO / PyTorch
Preprocess; return bounding boxes

Database Repositories
[Components]
Python / Psycopg
Transactional reads and writes

CUDA device access

SQL over TCP

NVIDIA GPU - optional
[Host resource]

Database [Container]
PostgreSQL 16

KEY
Amber: proxy; blue: routing
Teal: processing; grey: data; rose: hardware
Internal solid arrow: Python call
External solid arrow: network request
Thick arrow: hardware access
Enclosed blocks: components, not processes